§ Tensor product of Crystal (Notation in Book : (n-1)-r)

Main Examples

$GL_n(\mathbb{C})$ - Crystal or $B_{(1)}$

(Denoted $\mathbb{B}$ when n is understood)

Recall: For any seminormal crystal, it is determined by f_i ($i \in I$), that is, the crystal graph.

$(n-1)$-colors

$$\boxed{1} \xrightarrow{1} \boxed{2} \xrightarrow{2} \boxed{3} \cdots \boxed{n-1} \xrightarrow{n-1} \boxed{n}$$

ℓ_1

$GL_2(\mathbb{C})$ - Crystal $\quad \underline{1} \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \cdots \longrightarrow n$ ($L(n)$)

Defn: Given (seminormal) crystals $\mathcal{B}, \mathcal{C}$, define the **tensor product** seminormal crystals $\mathcal{B} \otimes \mathcal{C}$ with

As a set: Cartesian product $\{b \otimes c \mid b_i \in \mathcal{B},\ c \in \mathcal{C}\}$

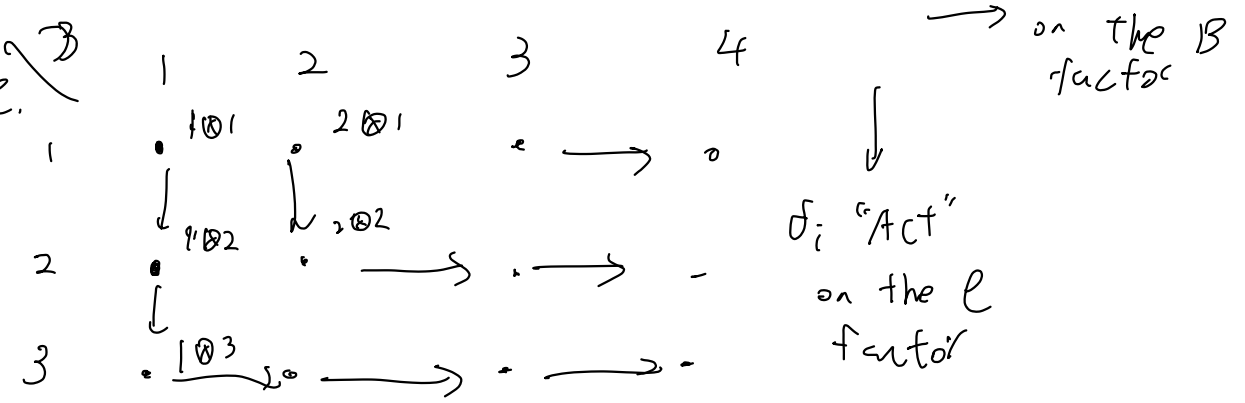
$$f_i(b \otimes c) = \begin{cases} f_i(b) \otimes c & \text{if } \varepsilon_i(b) \geq \varphi_i(c) \\ b \otimes f_i(c) & \text{otherwise.} \end{cases}$$

And $0 \otimes c = 0 \otimes b = 0$ for any b, c.

Ex: $\mathcal{B}$: $1 \to 2 \to 3 \to 4$ $\quad \otimes \quad$ $\mathcal{C}$: $1 \to 2 \to 3$

i	0	1	2	3
ε_i	0	1	2	3
φ_i	2	1	0	

e.g.



E.g. $\boxed{1} \xrightarrow{1} \boxed{2} \xrightarrow{2} \boxed{3} \xrightarrow{3} \boxed{4}$ $\qquad$ $\boxed{2} \xrightarrow{1} \boxed{2} \xrightarrow{2} \boxed{3} \xrightarrow{3} \boxed{4}$

ε_1	0	1	0	0	φ_1	1	0	0
ε_2	0	0	1	0	φ_2	0	1	0
ε_3	0	0	0	1	φ_3	0	0	1

And the element $1 \otimes 3$.

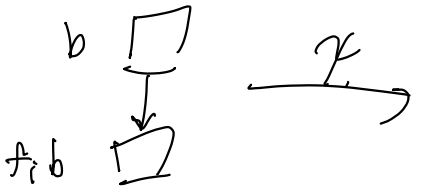
Next: Tensor product is associative. ... isomorphism?

(can assume seminormal)

Defn. Let $\mathcal{B}, \mathcal{C}$ be crystal of the same root system,

A $\mathcal{B} \to \mathcal{C}$ morphism. is a function $|\mathcal{B}| \to |\mathcal{C}| \cup \{0\}$

s.t. (1) wt, ε_i, φ_i are unchanged, and

(2) 

Same for e_i.

Remark. Sending ≥ 1 elements to 0 is allowed here.

Defn. $\Psi: \mathcal{B} \to \mathcal{C}$ morphism of crystal.

Ψ is strict if (2) happens for all $b \in \mathcal{B}$.

Ψ is an isomorphism if $\tilde{\Psi}: \mathcal{B} \sqcup \{0\} \to \mathcal{C} \sqcup \{0\}$ is bijective. $\tilde{\Psi}(b) = \begin{cases} \Psi(b) & \text{if } b \neq 0 \\ 0 & \text{if } b = 0 \end{cases}$

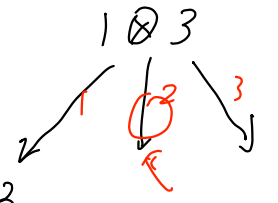
Thm. $(\mathcal{B} \otimes \mathcal{C}) \otimes \mathcal{D} \cong \mathcal{B} \otimes (\mathcal{C} \otimes \mathcal{D})$.

Thm. (2.14),

$$\varphi_i(x_1 \otimes x_2 \otimes \cdots \otimes x_k) = \max\left\{ \sum_{h=1}^{j} \varphi_i(x_h) - \sum_{h=1}^{j-1} \varepsilon_i(x_h) \;\middle|\; 1 \leq j \leq k \right\}$$

Moreover, if the maximum is attained at $j = \ell$.

$$f_i(x_1 \otimes \cdots \otimes x_k) = x_1 \otimes \cdots \otimes f_i(x_\ell) \otimes x_{\ell+1} \otimes \cdots \otimes x_k.$$



$f_1(1)\otimes 3 = 2\otimes 3$ $\quad 1\otimes 4 = 1\otimes f_1(3)$

Must go to zero since they act to 0 as well.

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§ Signature Rule for $B^{\otimes k}$.

Target, find, say $f_2(1\otimes 3\otimes 3\otimes 2\otimes 2\otimes 1\otimes 2\otimes 5)$

Given GL(n) crystal B, natural number k, $(x_1\otimes\cdots\otimes x_k)\in B^{\otimes k}$, $i\in I$

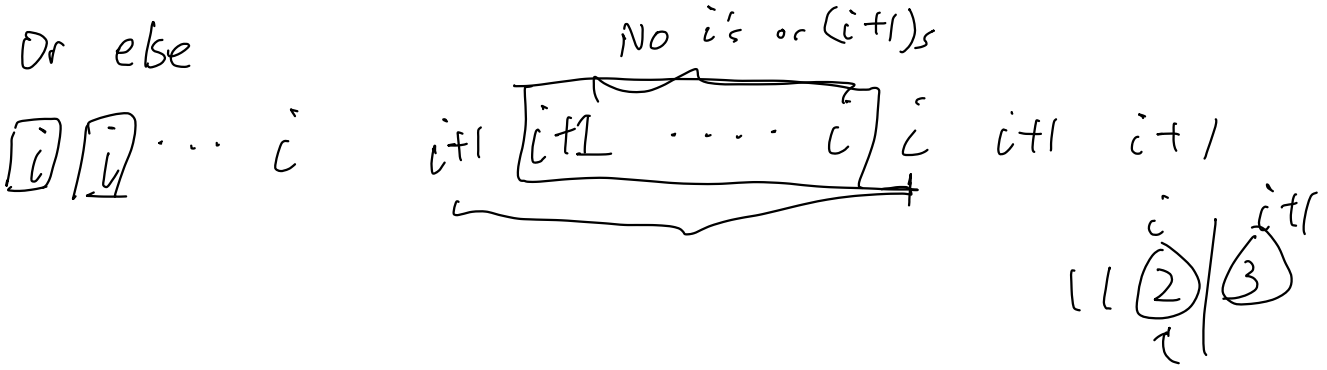
what is $f_i(x_1\otimes\cdots\otimes x_k)$?

$i\ i\ i\ i\ |\ i+1\ i+1\ i+1$

↑

Then f_i acts on this.

Or else



f_i acts on this.

$+\ +\ -\ -\qquad -\qquad +$

$1\ \underline{2}\ 3\ 1\ 1\qquad 2\ \underline{2}\ \underline{3}\ \underline{3}\qquad 1\ 2\ 1\ 1\ \underline{3}$

↑

Python verification.